Milestone-3 Report

Geospatial Analysis of US Natural Disaster Declarations

1. Introduction

Natural disasters such as hurricanes, floods, wildfires, and storms occur frequently across different regions of the United States. Understanding where disasters occur more often can help governments and organizations improve disaster management and preparedness.

The objective of this milestone is to perform **geographical analysis of disaster declarations** and visualize them using maps and charts. Geospatial visualizations make it easier to understand patterns that are difficult to observe in raw data tables.

This milestone focuses on identifying **disaster-prone states and geographical hotspots** using data visualization techniques.

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2. Problem Formulation

Analysis Type

- Exploratory Data Analysis (EDA)
- Geospatial Data Analysis

Key Questions

This analysis aims to answer the following questions:

1. Which states have the highest number of disaster declarations?
2. Are disasters concentrated in certain regions of the United States?
3. Do specific disaster types occur more frequently in certain states?
4. Can we identify geographical disaster hotspots?

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3. Dataset Description

The dataset used for this milestone is the cleaned dataset created in **Milestone-1**.

Input File

database.csv

The dataset includes the following columns:

Column	Description
State	US state where the disaster occurred
Disaster Type	Type of disaster (flood, hurricane, fire, etc.)
Declaration Date	Date when disaster was declared
Year	Extracted year from declaration date

These fields are important for analyzing disasters geographically and over time.

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4. Data Preparation

Before visualization, the dataset was cleaned and validated.

Steps Performed

1. Loaded the dataset using **Pandas**
2. Checked for missing values
3. Removed rows where the **State value was missing**
4. Converted **Declaration Date** into datetime format
5. Extracted **Year** from the declaration date
6. Grouped the data to calculate disaster counts by state

Example aggregation:

State	Disaster Count
Texas	High
California	High
Florida	High

This aggregation helps identify which states experience disasters more frequently.

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5. Tools and Libraries Used

The following Python libraries were used in the analysis.

Data Processing

- **Pandas**
- **NumPy**

Visualization

- **Matplotlib**
- **Seaborn**
- **Plotly**

Geospatial Visualization

- **GeoPandas**
- **Folium**

Plotly was mainly used for creating **interactive maps and animated visualizations**.

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6. Visualizations and Analysis

6.1 Top 10 States with Highest Disaster Declarations

Chart Type

Bar Chart

Description

This chart shows the **top 10 states with the highest number of disaster declarations**. The data was aggregated by grouping the dataset based on the state and counting disaster occurrences.

Insight

The visualization clearly highlights the states that experience disasters more frequently. States such as **Texas, California, and Florida** typically appear at the top due to their exposure to hurricanes, wildfires, floods, and storms.

These states are geographically vulnerable to different types of natural hazards.

6.2 Disaster Type Distribution by State

Chart Type

Stacked Bar Chart

Description

A stacked bar chart was created to analyze how different disaster types contribute to the total number of disasters in each state.

The data was grouped by:

- State
- Disaster Type

Then converted into a **pivot table** for visualization.

Insight

This visualization helps identify **which disaster types are dominant in specific states**.

For example:

- Coastal states may show more **hurricanes and storms**
- Western states may show more **wildfires**
- Some states may experience frequent **floods**

This helps understand the **disaster composition of each state**.

6.3 Choropleth Map of US Disaster Declarations

Chart Type

Geographical Choropleth Map

Description

A choropleth map was created using **Plotly** to visualize disaster counts across US states.

In this map:

- Each state is colored based on the **number of disaster declarations**
- **Darker colors represent higher disaster frequency**
- **Lighter colours represent fewer disasters**

Insight

The choropleth map clearly highlights **geographical disaster hotspots**.

It becomes easy to identify clusters of states that experience disasters frequently. Southern and coastal regions often appear darker because they are more exposed to hurricanes, floods, and tropical storms.

6.4 Animated Choropleth Map (Disasters over Time)

Chart Type

Animated Geographic Map

Description

An animated choropleth map created using Plotly to visualize how disaster declarations change **year by year**.

The animation frame is based on the **Year column**, which allows the map to show how disasters evolve over time.

Insight

This visualization helps observe:

- Changes in disaster frequency across years
- Growth or reduction in disaster declarations
- Trends in disaster patterns over time

Animated visualizations make it easier to understand **temporal changes in geographical disaster patterns**.

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7. Key Findings

From the analysis and visualizations, the following insights were observed:

1. Certain states experience significantly more disasters than others.
2. Coastal and southern states tend to have higher disaster declarations.
3. Different states experience different dominant disaster types.
4. Geographical maps make it easier to identify disaster-prone regions.
5. Disaster patterns change over time and can be visualized effectively using animated maps.

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8. Conclusion

This milestone focused on analyzing and visualizing the **geographical distribution of natural disasters in the United States**.

Using geospatial visualization techniques such as **choropleth maps, bar charts, and stacked charts**, the analysis revealed important spatial patterns and disaster hotspots.

These visualizations help transform raw data into meaningful insights that can support **disaster planning, risk management, and policy decisions**.

Geospatial analysis is a powerful tool for understanding how disasters affect different regions and how disaster patterns evolve over time.

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